The Shades — FOH EQ Rationale

Fountain Square (outdoor) · DiGiCo Quantum 225 · 2026-08-02 · Rev 1.0 Deep Build

THE ARTIST — AND WHAT IT MEANS FOR THIS MIX

Mic. Carr & Shades — a Cincinnati band billed on the input list as The Shades, formally Michael 'Mike' Carr and the Shades. Their own copy calls them a live-instrument soul and alternative R&B band founded by songwriter and artist Mic. Carr, blending thoughtful storytelling with warm, organic instrumentation, and describes the live experience as less like a performance and more like a warm, shared meditation. That is a band description that tells you how to mix them: organic, low-mid forward, un-scooped, vocal-centred. Carr is the songwriter, the namesake and the male lead voice; the band is credited with smooth harmonies, which is what the two additional wireless channels are carrying. Saxophonist Elijah Woodward joined for a single show and stayed, per the Spectrum News feature on Carr's decade-anniversary album Broken Crayons Still Color — he is CH 19, and in this genre the horn sits inside the arrangement rather than on top of it. They play originals from the 2026 Shades EP alongside classic covers, so expect the dynamic range of a covers set with the tonal palette of a neo-soul record. Practical read for the desk: protect warmth, keep the vocal forward and the horn folded in, and resist the urge to make the kit bright.

THE ROOM — AND THE CONDITIONS ON THE NIGHT

Fountain Square, open plaza, no room gain and no natural late field — and 87-94% relative humidity for the whole show window. Two rules collide here and the reconciliation is on paper deliberately. The FSQ depth rule says cuts run -6 to -9, deeper than indoors; the genre says low-mid warmth is the point of the band. Both are true because outdoor mud is MECHANICAL — drum shell box, guitar cab resonance, stage bleed — rather than room buildup, since a plaza has none. So the full outdoor depth is spent where the mud is actually made: -8 on the kick's 400 Hz box, -7 on the snare, the hat, the overheads and the guitar cab. It is deliberately withheld where the low-mids are the instrument: -5 on the keyboards' 300 Hz and -5 on the vocals' 700 Hz. Humidity then sets the top end in one direction across every channel — saturated air passes HF to the back of the plaza essentially intact, so no channel in this build carries an HF boost, every baked presence peak is trimmed, and the hat's usual 8-10 kHz shimmer lift is inverted into a cut. Wind is negligible at 1-4 mph, so the overheads get HPF 300 rather than a windy night's 400.

What changed from the KB default / prior rev — and why

- Locker fork CH 3 snare top — swapped the specified Sennheiser e604 for the Audix i5. Brian delegated the call. The e604's documented thinness disappears outdoors with no room gain; the i5's +5 dB at 150 Hz is the body a soul snare needs, and its one liability at 5.5 kHz was getting trimmed on a humid night regardless. Knock-on: only two e604s needed now, both on racks.
- Locker fork CH 19 sax — the specified EV N/D 408 and my Beta 98H/C alternative were both declined; Brian chose the Audio-Technica PRO 35.
- CH 9/10 — the input list put OH L on 9 and OH R on 10. Overridden: FSQ CH 10 is the SNARE PL8 return, so the pair runs STEREO on fader 9. Same as the last two FSQ builds.
- CH 2 / CH 8 — the input list read '52' on kick out and 'B52' on the floor tom, which would need two Beta 52As. Brian resolved it: Audix D6 on kick out, Beta 52A on the floor.
- CH 33-35 — the FSQ template's baseline HPF of 184 Hz is overridden on all three vocal faders (90 on the lead, 110 on the harmonies). 184 sits above the entire lower octave of any male voice.
- Hat CH 5 — the genre-standard 8-10 kHz shimmer lift is inverted into a -3 cut at 9 kHz. Driven by the fetched 87-94% RH, not by taste.
- Keys CH 17/18 and vocals CH 33-35 — low-mid cuts held at -5, short of the FSQ -8. Genre outranks the venue rule here and the reasoning is in room_context so it is auditable rather than looking like an oversight.
- Floor tom CH 8 — outdoor depth also withheld (-3/-5 rather than -7/-8), but for a capsule reason rather than a genre one: the Beta 52A's low-mids are already scooped.
- CH 4, 14, 15, 16 and 20-32 — dropped. Per Brian: no snare bottom, one guitar only, and the Misc rows were template leftovers.

Question round — what was asked, what Brian decided

- Q: is this one bill or two shows? -> Brian: two shows. Separate folders, packets and .ses files.
- Q: how many vocals and on what? -> Brian: three wireless for both acts, and that's all.
- Q: snare bottom, guitars 2-4, keys DIs, Misc 20-24? -> Brian: no snare bottom, one guitar, IMP DIs on the keys, 20-24 dead.
- Q: two Beta 52As on the list — which channel keeps it? -> Brian: D6 on kick out, Beta 52A on the floor tom.
- Q: Ric Sexton's sax — pedal feed only, and does he split alto/soprano? -> Brian: he swaps horns on the one channel and brings his own mic.
- Locker fork CH 19 sax: N/D 408 specified, Beta 98H/C offered -> Brian: neither, use the Audio-Technica PRO 35.
- Locker fork CH 3 snare top: e604 specified, Audix i5 offered -> Brian: 'you choose the better option, I'll follow' -> swapped to the i5, reasoning recorded in changes and mic_notes.
- Q: vocalist voice types for the wireless slotting? -> Brian: doesn't know, asked for a researched suggestion. Research could not resolve them either, so the three faders are built as role slots with distinct cut lanes and a documented swap procedure.
- Spelling: Ric Sexton, not Rick — confirmed by Brian.

Reverb — Seventh Heaven Pro (100% wet returns)

• **VOCAL** — **Chambers 1 / #10 Vocal Chamber** | Settings: Decay 1.40s (from 1.60s factory) · PreDelay 20ms (from 0 factory) · VLF -14 (from -10 factory) · E/L Max Early · Late -6 · Late Rolloff 8000 (factory) | In-plugin EQ: LC 200 Hz on the return; ducker in Reverb mode, threshold -22 dB, release 300 ms | Why: The default for Mic. Carr and the one most likely to be up all night. Soul and neo-soul want a reverb that is warm and placed rather than bright and long, and a V1 chamber gives him somewhere to stand without gloss. Decay comes down from factory because an open plaza offers no late field of its own and no walls to justify a long tail. Pre-delay goes to 20 ms from the factory zero to keep his consonants intact over a band, and VLF is pulled from -10 to -14 so his chest register stays out of the tail, where at 90% humidity it would otherwise thicken.

• **VOCAL** — **Plates 1 / #06 Vocal Plate** | Settings: Decay 1.50s (factory) · PreDelay 24ms (factory) · VLF -19 (factory) · E/L Max Early · Late -8 · Late Rolloff 6400 (factory) | In-plugin EQ: LC 180 Hz on the return; ducker Reverb mode, threshold -22 dB | Why: The up-tempo option — the covers in the set that need the vocal dense and forward rather than placed in a space. Everything sits at factory here because the factory values are already right for outdoors: 24 ms of pre-delay is enough separation over a band, and VLF at -19 is dark enough that nothing muddies. Late runs slightly lower than the chamber's so the two can be blended without doubling the tail.

• **VOCAL** — **Plates 2 / #13 Vocal Shimmer** | Settings: Decay 1.60s (from 1.80s factory) · PreDelay 20ms (factory) · VLF -14 (factory) · E/L Max Early · Late -6 · Late Rolloff 8000 (factory) | In-plugin EQ: LC 220 Hz on the return; ducker Reverb mode, threshold -20 dB, release 400 ms. The preset's baked 236 ms delay is part of the shimmer — leave it. | Why: The big-moment option, and this band earns one: their own material is written around resilience and reflection, and a covers set will have at least one ballad where Carr is out front alone. V2 bloom rather than V1 realism, which is the right character for a modern R&B lead. Decay pulled a step under factory because outdoors the tail has nothing to reflect off, but the rolloff stays factory-bright at 8 kHz — there is no audience HF absorption out there to compensate for.

• **INSTRUMENT** — **Plates 1 / #04 Snare Plate A** | Settings: Decay 1.00s (from 1.20s factory) · PreDelay 0 (factory) · VLF -12 (from -9 factory) · E/L Max Early · Late -8 · Late Rolloff 6800 (factory) | In-plugin EQ: LC 250 Hz on the return to keep the kit's low-mids out of the plate | Why: This is the plate that lives on the template's fader 10 return, so it costs nothing to have ready. A soul backbeat with ghost notes wants a short, dense plate that makes the snare feel bigger without smearing the ghosts — hence decay pulled to 1.0s and VLF cut a little further than factory so the plate adds width rather than weight.

• **INSTRUMENT** — **Rooms 1 / #10 Large Wooden** | Settings: Decay 1.00s (from 1.20s factory) · PreDelay 0 (factory) · VLF -12 (from -9 factory) · E/L Max Early · Late -8 · Late Rolloff 8000 (factory) | In-plugin EQ: LC 200 Hz on the return; leave the rolloff factory-bright | Why: For the sax and the keyboards. A bell-clipped PRO 35 hears a very close, very dry horn, and outdoors there is nothing at all to put air around it — a short wooden room is the smallest honest fix, and it works equally on a Rhodes that would otherwise sit flat against the vocal. Kept under a second so it reads as air rather than as an effect.

• **GENERAL** — **Rooms 1 / #02 Studio B Close** | Settings: Decay 0.75s (factory) · PreDelay 0 (factory) · VLF -12 (from 0 factory) · E/L Max Early · Late -10 · Late Rolloff 8000 (factory) | In-plugin EQ: LC 300 Hz on the return. VLF must come down from its 0 dB factory value — this is the preset the KB flags for exactly that. | Why: Drum-bus glue at a very low send, to stop the kit sounding like eight separate close mics stapled together on a stage with no room. Factory decay is already short enough; the only real move is the VLF, which ships at 0 dB and would otherwise put weight back into a low end that is already carrying a kick pair and a bass pair.

• **USING THEM TOGETHER** — Vocal Chamber is the bed and should be the one that is up by default — warm, placed, and unobtrusive enough to leave on through a whole set. Vocal Plate is its up-tempo counterpart: when the band lifts into a cover and Carr needs to be dense and in front rather than set back in something, cross to the plate instead of pushing the chamber, because pushing a chamber outdoors just makes it audible as a room that isn't there. Vocal Shimmer is a moment, not a setting — bring it up for one ballad and take it back down. Never run two vocal options up at once; their tails are all in the same 1.4-1.6 second range and they will simply sum into wash. Underneath, Snare Plate A and Large Wooden are doing different jobs at different frequencies — the plate is width above 250 Hz on one drum, the wooden room is air above 200 Hz on the horn and keys — so they coexist without stacking. Studio B Close sits under everything at a genuinely low send and is the first thing to pull if the mix starts to feel soft; its LC at 300 Hz is what keeps it from becoming mud.

RESEARCH — WHAT WAS LOOKED UP, AND WHAT IT CHANGED

GENRE — VERIFIED FIRST, WITH NAMED EVIDENCE	THE GIG	CONDITIONS (FETCHED, NEVER ASSUMED)
Live-instrument soul / alternative R&B with jazz and funk underneath. Verified BEFORE any other research against named evidence: the band's own description on CincyMusic ('a live-instrument soul and alternative R&B band founded by songwriter and artist Mic. Carr'), shadestheband.com, and the Spectrum News feature of 2025-08-15 covering Carr's Broken Crayons Still Color and naming saxophonist Elijah Woodward in the band. Evidence is consistent across all three — not split, so no stop-and-ask was triggered.	Fountain Square, 2026-08-02. One of two separate shows Brian is building for the same night at the same venue on a shared backline; Ric Sexton is the other. Separate folders, separate packets and separate .ses files per Brian's call. 17 active channels: full kit, bass DI plus cab, one guitar, a stereo keyboard rig, one sax and three house wireless vocals.	Open-Meteo, pulled 2026-08-02 for 39.1015/-84.5125, show window 5-11pm: 71-75 F, relative humidity 87-94%, wind 1-4 mph gusting to 9, rain 15-25% easing to 4% late. Fetched, not assumed. Saturated air is the headline: at 87-94% RH the top end reaches the back of the plaza essentially intact, so no channel in this build carries an HF boost and every capsule's baked presence peak is trimmed instead. Light wind puts the overhead pair at HPF 300 rather than the 400 a gusty night earns.

CH	SOURCE / MIC	WHAT THE RESEARCH FOUND	V	TRACE — base · equip · genre · artist · venue
1	**Kick in** Shure Beta 91A	Two-step contour switch centred at 400 Hz attenuating about -7 dB; boundary mount, pronounced beater click, picks up 300-500 Hz shell boxiness. *Thomann / Shure Beta 91A spec; Mix Online 'Shure Beta91A real-world review'*	AGREE	**base** Boundary kick mic — 400 Hz shell box is the capsule's own documented problem, cut -8 there **equip** no drum sizes or head types notated — no change **genre** soul kick wants weight and a felt beater, so the pronounced click is trimmed -3 at 5 k rather than left **artist** organic, un-scooped band sound — no modern click voicing added **venue** FSQ outdoor — box cut taken to the full -8; HPF raised to 60 to vacate the D6's low lane
2	**Kick out** Audix D6	+14 dB at 60 Hz, -15 dB scoop at 700-750 Hz, +15 dB at 4-5 kHz, +17 dB at 10-12 kHz; 30 Hz-15 kHz. *Audix AX-D6 spec sheet; RecordingHacks D6 curve*	AGREE	**base** pre-voiced smiley curve — B1 and B2 left FLAT because the boost and the scoop are both already baked **equip** no kick size notated — no change **genre** the +15 at 4-5 k and +17 at 10-12 k are a rock voicing; soul kick wants neither, so LPF 7 k kills the top peak and B4 trims the other **artist** no programmed kick in this band, so the live kick carries the whole low end — HPF down at 35 **venue** FSQ — no room gain, so the low end is passed through rather than trimmed
3	**Snare top** Audix i5	+5 dB at 150 Hz and +9 dB at 5500 Hz; -3 dB points at 50 Hz and 16 kHz; 1.6 mV/Pa sensitivity, which measurably limits hi-hat spill. *RecordingHacks Audix i5; Sound On Sound Audix i5 review*	AGREE	**base** locker fork resolved to the i5 — B1 left FLAT because the +5 at 150 Hz body lift is baked **equip** no snare size or head notated — no change **genre** soul snare is fat and round, not cracky, so the +9 at 5.5 k is trimmed -4 rather than exploited **artist** ghost notes carry the groove — any gate stays shallow at -8 range so they live **venue** FSQ — box cut to -7, but HPF held at 120 not 180 so the outdoor rule doesn't eat the capsule's body lift
5	**Hi-hat (under-mounted)** Shure SM81	Ruler-flat 20 Hz-20 kHz with a switchable LF filter at 6 or 18 dB/oct and a -10 dB pad — nothing voiced in. *Shure SM81 specification; B&H product data; soundref SM81 review*	AGREE	**base** flat capsule — the template applies straight, no correction needed **equip** no hat model notated — no change **genre** the genre default 8-10 kHz shimmer lift is INVERTED to a -3 cut — see the venue layer **artist** organic band sound, no hyped top — confirms the inversion **venue** FSQ at 87-94% RH — HF arrives intact, so 9 kHz is cut not lifted; under-mount means more shell bleed, so 500 Hz takes the full -7

CH	SOURCE / MIC	WHAT THE RESEARCH FOUND	V	TRACE — base · equip · genre · artist · venue
6/7	**Rack toms** Sennheiser e604	40 Hz-18 kHz, 160 dB max SPL; measurably thin on rack toms and wants a low boost, presence sitting in the upper mids rather than the extreme top. *Sennheiser e604 product specification; Gearspace and HomeRecording rack-tom threads*	AGREE	**base** documented low-mid scoop — licenses the only low boost in the drum section, +4 dB **equip** no tom sizes notated — boosts split 180/150 Hz by position instead **genre** soul toms want body and pitch, so the voiced attack is trimmed -3 **artist** no change — nothing act-specific about the toms in either band **venue** FSQ — box cuts at -6 on both
8	**Floor tom** Shure Beta 52A	Tailored kick voicing: presence lift around 4 kHz, scooped low-mids, big lows; 'a very EQ'd and scooped tone with a huge low end up close.' *Shure Beta 52A spec via Thomann/FrontEndAudio; Drum Forum (DFO) and TalkBass floor-tom threads*	AGREE	**base** kick mic on a floor tom — B1 FLAT because the big lows are baked **equip** no floor tom size notated — no change **genre** soul floor tom is deep and round, so the 4 kHz kick-attack lift is trimmed -4 **artist** no change **venue** FSQ depth deliberately NOT taken — B3 -3 and B2 -5 rather than -7/-8, because the forums' one warning is that this mic gets lost in a dense mix and the low-mid scoop is already baked
9	**Drum overheads (stereo)** Shure Beta 27 (pair)	Nominally flat 60 Hz-3 kHz with two small HF peaks of +2 dB or less at 5500 Hz and 9000 Hz; HF -3 dB point above 15 kHz; pattern consistent below 6400 Hz. *RecordingHacks Beta 27 measurement; Shure Beta27 spec sheet*	AGREE	**base** both bands land exactly on the two measured capsule peaks — trims, not lifts **equip** no cymbal detail notated — no change **genre** overheads carry the kit's glue in a soul mix, not just cymbals, so they aren't treated as a cymbal-only channel **artist** no change **venue** wind 1-4 mph gusting 9 — HPF 300 not the gusty-night 400; 87-94% RH — both peaks trimmed -3; stage bleed cut takes full -7
11	**Bass DI** Whirlwind IMP (passive DI)	No capsule — the researched fact is the blend rule: the DI works best below 100 Hz, the mic defines tone above 100 Hz, and splitting them there removes the phase problem entirely. *TalkBass 'Blending DI + Mic Live?' consensus; Sound On Sound 'Matching The Phase Of Mic & DI Signals'*	THIN	**base** DI owns everything below 100 Hz — +3 at 60 Hz, gate-cleared because there is no baked curve and the cab mic is HPF'd at 100 **equip** no amp, cab or string type notated — no change **genre** round fingerstyle pocket, no grind — nothing added above **artist** no change **venue** FSQ — no room gain, so the sub lane is supported rather than trimmed
12	**Bass cabinet** Shure PG52	30 Hz-13 kHz; published curve humps at 60-100 Hz, dips broadly through 200-800 Hz, rises at 4-5 kHz, rolls off hard above 10 kHz. *Shure PG52 specification sheet; TalkBass 'Shure PG-52 for miking bass amp?'; RecordingHacks*	AGREE	**base** three capsule-gate calls — B1 flat (baked hump), B2 at 160 not 300 (baked dip), B4 trims the 4-5 kHz rise **equip** no cab model notated — no change **genre** warm round bass tone — the 4-5 kHz rise reads as clank, not definition, so it comes off **artist** no change **venue** FSQ — HPF 100 doubles as the lane boundary; box cut -5 at the KB-prescribed 160 Hz
13	**Guitar cabinet** Sennheiser e609 Silver	40 Hz-15 kHz supercardioid with a presence peak at about 4 kHz and a broader midrange peak across 3-6 kHz. *Sennheiser e609 Silver product specification; homestudiobasics e906/e609 comparison; neo-soul staging guidance (cut guitar 2.5-3 kHz to clear the vocal)*	AGREE	**base** both bands sit inside baked peaks — both are trims **equip** no amp or cab model notated — no change **genre** the 2.8 kHz vocal-clearing cut comes straight from the neo-soul source, deepened past its 1-2 dB because outdoors nothing blurs the overlap **artist** one guitar carrying all the harmony — held to -3 rather than -5 so it isn't gutted **venue** FSQ — cab box takes the full outdoor -7, which is exactly the mechanical mud the rule targets

CH	SOURCE / MIC	WHAT THE RESEARCH FOUND	V	TRACE — base · equip · genre · artist · venue
17/18	**Keyboards (stereo rig)** Whirlwind IMP (passive DI)	No capsule. The carrying fact is genre-sourced: neo-soul staging guidance is to reduce 8-10 kHz by 3-5 dB against stage reverb and to treat ~500 Hz proximity warmth as a feature to protect. *Neo-soul production/staging guidance (Stealify, Grokipedia neo-soul, ClefArc); Whirlwind IMP specification*	THIN	**base** neutral passive DI — every move is a separation decision, no correction needed **equip** no board models notated — Rhodes/electric piano assumed, flagged in notes **genre** 8 kHz trimmed per the source; 2 kHz bark cut to keep the boards off the vocal **artist** warm organic instrumentation is the band's own self-description — confirms protecting the low-mids **venue** FSQ depth HELD at -5 not -8 — a Rhodes' 300 Hz is the instrument, and an open plaza adds no buildup to it
19	**Saxophone** Audio-Technica PRO 35	50 Hz-15 kHz with a built-in LF roll-off at 80 Hz at 18 dB/oct, 145 dB max SPL, subtly rounded treble. Instrument fact: alto fundamentals sit 150-700 Hz with presence at 2-4 kHz. *Audio-Technica PRO35 product specification / Thomann; sax live-mixing guidance (Sax on the Web, musicalinstrumenthub live EQ guide)*	AGREE	**base** clip mic on the bell — HPF set at 120, above the capsule's own 80 Hz roll-off so the two don't duplicate **equip** no horn or mouthpiece notated — no change **genre** neo-soul horn sits inside the arrangement — the 2.5 kHz honk cut does the work, nothing is added on top **artist** Elijah Woodward plays inside the band rather than over it — no solo-forward presence lift **venue** FSQ — bell-proximity box cut at -6; rounded treble plus 87-94% RH means an air lift would be wrong twice over
33-35	**Vocals (lead + two harmonies)** Shure Beta 58A (house wireless)	50 Hz-16 kHz, attenuated below 500 Hz to counter proximity, with TWO presence peaks at 4 kHz and 10 kHz. Vocal-range references: male live HPF 60-100 Hz, female 80-120 Hz. Voice types could not be resolved: no published source classifies these singers and Brian doesn't know them, so that layer is THIN and is handled as a soundcheck swap rather than a guess. *Shure Beta 58A specification sheet; Sound On Sound Beta Series review; vocal-range/HPF references carried from the 2026-07-31 2nd Wind build*	AGREE	**base** cuts only, every band; B4 and B3 land on the two measured capsule peaks **equip** house wireless, fixed rig — no change **genre** vocal-forward soul with smooth harmonies — warmth protected, separation taken in the nasal lane instead **artist** Mic. Carr is the male lead and namesake, so CH 33 keeps the most 4 kHz (-2, tight Q3) and stays the most intelligible voice **venue** template HPF 184 OVERRIDDEN — 90 on the lead, 110 on the harmonies; 87-94% RH means both baked peaks are trimmed on all three

Reconciliation — every web vs KB fork, and how it was settled

• Audix D6: the web puts the mid scoop at 700-750 Hz where the KB says ~600 Hz. Same feature, and the web figure is the more precise one — taken as the working value and logged for KB write-back. The practical consequence is real: B3 was moved to 1200 Hz rather than cutting into the scoop.

• Audix i5 vs Sennheiser e604 on snare top: no disagreement between web and KB — both describe the i5 as body-lifted at 150 Hz and the e604 as thin and scooped. The fork was a judgement call Brian delegated, not a source conflict.

• Bass DI + cab blend: the KB has no entry for this at all, so the lane split is web-only. Verdicted THIN rather than papered over as agreement.

• Keyboards on a DI: the KB has no keyboard or line-level row — the same gap logged on the 2026-07-27 and 2026-08-01 builds. Verdicted THIN, values kept conservative.

• Vocal voice types: could not be resolved by research. Brian doesn't know them and no published source classifies these singers. Built as role slots with a male-and-female-safe HPF and three distinct cut lanes, and flagged for a wholesale curve swap at soundcheck rather than guessed at.

KB write-back candidates — no article covers these yet

• Audix D6 — update the mid-scoop centre from ~600 Hz to the measured 700-750 Hz, and add the +17 dB at 10-12 kHz figure the KB row currently lacks.

• Bass DI + cab blend — no KB entry exists. Propose a mic-library blend row: DI owns below 100 Hz, cab mic owns above, HPF the cab at 100 as the lane boundary, mono-sum and flip the cab if thinner.

• Keyboards / line-level boards — still no KB row after three builds. Propose an eq-starting-points section.

• Shure Beta 52A on floor tom — the 'monster on floor toms once the kick mic is something else' consensus, plus the do-not-stack-the-outdoor-cut-on-a-baked-scoop rule, is worth a mic-library note.

• Audio-Technica PRO 35 on sax — the built-in 80 Hz 18 dB/oct roll-off isn't in the KB row, and it changes where the desk HPF should sit.

DRUMS

Ch 1 | Kick In · Shure Beta 91A

HPF 60 | LPF 8000 || B4 -3@5 kHz Q2 BELL   B3 -8@400 Hz Q2 BELL   B2 -6@250 Hz Q1.8 BELL   B1 flat

The 91A's two-step contour switch is centred at 400 Hz and pulls about -7 dB when engaged (Thomann/Shure spec via the Mix Online review), and the KB names 300-500 Hz shell boxiness as its standing weakness. Same fact from two directions. The switch is assumed FLAT, so the 400 Hz cut lives on the desk — if you find it engaged at load-in, halve the B3 cut to -4. Nothing is boosted here: this is the attack half of the pair and it deliberately vacates the bottom.

Two-mic kick, and this mic owns the attack and mid-definition lane top to bottom while the D6 on CH 2 owns the low and the body. That is why the HPF sits up at 60 and B1 is flat — the D6 already bakes in +14 dB at 60 Hz, and a second boost there is the classic stacked-low failure. The beater click the 91A is known for gets trimmed rather than left, because a soul kick wants weight and a soft felt beater, not a metal click. The 400 Hz box cut takes the full outdoor -8: that is mechanical box resonance in a drum shell, exactly what the FSQ depth rule is for. GATE if you want one: threshold around -30, 5 ms attack, 150 ms release, shallow -10 range — documented, not patched.

Ch 2 | Kick Out · Audix D6

HPF 35 | LPF 7000 || B4 -5@4.5 kHz Q2 BELL   B3 -4@1.2 kHz Q2 BELL   B2 flat   B1 flat

The D6 is the most pre-voiced mic in the show: +14 dB at 60 Hz, a -15 dB scoop at 700-750 Hz, +15 dB at 4-5 kHz and +17 dB at 10-12 kHz (Audix AX-D6 spec sheet / RecordingHacks). The KB puts the scoop nearer 600 Hz; the web number is more precise and is logged as a KB write-back. Three bands are left flat on purpose and each one is a capsule-gate call: B1 because +14 dB at 60 is already there, B2 because 90-600 Hz is already attenuated, and B3 sits at 1200 rather than the obvious 700 because cutting into a baked -15 dB scoop is exactly the mistake the gate exists to catch.

This channel is mostly the capsule doing the work. The two moves that matter both pull the D6's rock voicing back toward soul: the LPF at 7 kHz removes the +17 dB at 10-12 kHz outright — pure beater fizz that a soul kick has no use for, and on a 90% humidity night it would otherwise carry all the way to the back of the plaza — and B4 trims the +15 dB attack peak at 4.5 kHz. Everything below stays untouched so this mic can own the weight lane cleanly under CH 1.

Ch 3 | Snare Top · Audix i5

HPF 120 | LPF 14000 || B4 -4@5.5 kHz Q2 BELL   B3 -3@1 kHz Q2 BELL   B2 -7@450 Hz Q2 BELL   B1 flat

The i5 carries +5 dB at 150 Hz and +9 dB at 5500 Hz, with -3 dB points at 50 Hz and 16 kHz and a low 1.6 mV/Pa sensitivity (RecordingHacks / Sound On Sound). That low sensitivity is specifically credited with keeping hi-hat spill off the snare channel, which matters here because the hat mic is mounted under the hats and is already hearing more shell than usual. B1 is flat because the +5 dB at 150 Hz is baked — the body on this channel is bought with the HPF corner, not with a band.

The fork went to the i5 over the specified e604 because the e604's documented weakness is being thin and boxy in the low-mids, and outdoors with no room gain a thin snare simply vanishes. The HPF is held at 120 rather than the usual 180 so the capsule's own 150 Hz body lift survives — the outdoor rule does not get to eat the body. Its one liability, the +9 dB at 5.5 kHz, is trimmed -4, which is a move this humid night wanted anyway, so the swap costs nothing tonight. The 450 Hz box cut takes the full outdoor -7. GATE philosophy if you gate it: shallow range, no more than -8, 2 ms attack, 120 ms release — ghost notes and press rolls are the feel in both of these bands and they have to live.

Ch 5 | Hat · Shure SM81

HPF 400 | LPF 16000 || B4 -3@9 kHz Q2 BELL   B3 -6@1 kHz Q2 BELL   B2 -7@500 Hz Q2 BELL   B1 flat

Ruler-flat 20 Hz-20 kHz with a switchable LF filter at 6 or 18 dB per octave and a -10 dB pad (Shure spec / B&H). Nothing is voiced in, which is the point — every move here is a decision rather than a correction, and there is no baked peak for anything to stack on.

The interesting band is B4, and it is a cut. The genre default on a hat is an 8-10 kHz shimmer lift; at 87-94% relative humidity the air passes the top end to the back of the plaza essentially intact, so that lift becomes a -3 at 9 kHz instead. That inversion is the humidity call made visible, and it is the same call the 96% RH show on 8/1 landed on. The 500 Hz cut takes full outdoor depth because an under-mounted hat mic sees the snare shell from below and that bleed is pure box.

Ch 6 | Rack 1 · Sennheiser e604

`HPF 90 | LPF 12000 || B4 -3@6 kHz Q2 BELL   B3 -4@700 Hz Q2 BELL   B2 -6@350 Hz Q2 BELL   B1 +4@180 Hz Q1.2 BELL`

40 Hz-18 kHz, 160 dB max SPL, fast transient response from a lightweight voice coil (Sennheiser spec sheet). The relevant finding is behavioural and the web and the KB agree on it: the e604 is thin on rack toms and wants a low boost — Gearspace and HomeRecording both land there, and the KB calls it scooped in the low-mids with modest low end. That agreement is what licenses the only low boost in the drum section: +4 dB at 180 Hz fills a hole the capsule cut, rather than stacking on a peak.

Round and pitched rather than attacking — soul toms are musical, not smashed, so the voiced attack gets trimmed at 6 kHz and the body gets put back underneath. The 350 Hz box cut runs -6 for the outdoor rule. Rack 1's body sits at 180 Hz and Rack 2's at 150 so the two toms stay off each other's fundamental.

Ch 7 | Rack 2 · Sennheiser e604

`HPF 80 | LPF 12000 || B4 -3@5.5 kHz Q2 BELL   B3 -4@600 Hz Q2 BELL   B2 -6@320 Hz Q2 BELL   B1 +4@150 Hz Q1.2 BELL`

Same capsule and same documented low-mid scoop as CH 6 (Sennheiser spec; Gearspace/HomeRecording tom threads), so the same +4 dB gate-cleared boost applies — moved down to 150 Hz for the lower drum.

The lower of the two racks, so everything shifts down a step: HPF to 80, the honk cut to 600 Hz, the box cut to 320 and the body boost to 150. Keeping the two toms' boosts a third apart is what stops a fill turning into one thick note outdoors.

Ch 8 | Floor · Shure Beta 52A

`HPF 50 | LPF 10000 || B4 -4@4 kHz Q2 BELL   B3 -3@800 Hz Q2 BELL   B2 -5@250 Hz Q2 BELL   B1 flat`

A kick mic on a floor tom, deliberately. Tailored kick voicing: presence lift around 4 kHz, scooped low-mids, big lows (Shure spec via Thomann; the KB concurs), and the forums describe it as 'a very EQ'd and scooped tone with a huge low end up close.' The DFO consensus is that it is an absolute monster on floor toms once the kick mic is something else — which is exactly the configuration this show runs. B1 stays flat because those big lows are already baked.

This is the one drum channel where the outdoor depth rule is deliberately NOT taken, and the reason is the capsule rather than the genre. B3 sits at -3 and B2 at -5, short of the -7/-8 the FSQ rule would give, because the mic already scooped its own low-mids and the single warning that repeats across every forum thread about it is that it gets lost in a dense mix. Stacking an outdoor cut on a baked scoop is how that happens. The 4 kHz presence lift is a kick-attack feature a floor tom has no use for, so it comes off -4.

Ch 9 | Overheads · Shure Beta 27 (pair)

`HPF 300 | LPF 16000 || B4 -3@9 kHz Q2 BELL   B3 -3@5.5 kHz Q2 BELL   B2 -7@400 Hz Q2 BELL   B1 flat`

Nominally flat from about 60 Hz to 3 kHz with two small HF peaks of +2 dB or less at 5500 Hz and 9000 Hz, and the HF -3 dB point above 15 kHz (RecordingHacks measurement / Shure Beta27 spec sheet). The supercardioid pattern stays consistent below 6400 Hz and narrows above it. B4 and B3 land exactly on those two measured peaks — both trims, no lift anywhere.

Wind is 1-4 mph gusting to 9, which is benign, so the pair takes HPF 300 rather than the 400 a gusty night would force. Humidity does the opposite job: at 87-94% RH both capsule peaks arrive intact at the back of the plaza, so they get pulled -3 each instead of being left to sparkle. In both of these bands the overheads carry the kit's glue and not just the cymbals, so the 400 Hz cut is aimed at stage bleed and box, not at the drums themselves, and it takes the full outdoor -7.

RHYTHM

Ch 11 | Bass DI · Whirlwind IMP (passive DI)

`HPF 30 | LPF 8000 || B4 flat   B3 -4@1.2 kHz Q2 BELL   B2 -5@400 Hz Q2 BELL   B1 +3@60 Hz Q1.2 BELL`

A passive DI has no capsule to characterise, so the researched fact carrying this channel is the blend rule itself: below 100 Hz the DI works best because it sidesteps the low-end limits of the cab and the mic, while tone is defined above 100 Hz where the mic captures the amp — use the DI only below 100 and the mic only above it and the phase problem disappears (TalkBass consensus, corroborated by Sound On Sound's 'Matching The Phase Of Mic & DI Signals'). The KB has an IMP row but nothing on bass DI-plus-cab blending at all, so this unit is verdicted THIN and the lane split is carried entirely by the web pass. The +3 dB at 60 Hz clears the gate because there is no baked curve to stack on and the cab mic is high-passed at 100 with B1 flat, so nothing else is boosting down there.

The DI owns the bottom octave and nothing else. It is low-passed at 8 kHz and cut at 400 and 1200 Hz specifically to vacate the midrange rather than compete in it, which is what keeps the two bass channels from combing. Neither band wants grind out of the bass — both live on a round, fingerstyle pocket — so the weight goes in here and the character comes off the cab. COMP: Mustard Blue (Neve), 4:1, 15 ms attack, 120 ms release, 3-4 dB of gain reduction — documented, not patched.

Ch 12 | Bass Cab · Shure PG52

`HPF 100 | LPF 9000 || B4 -4@4.5 kHz Q2 BELL   B3 -3@1.5 kHz Q2 BELL   B2 -5@160 Hz Q2 BELL   B1 flat`

30 Hz-13 kHz, a cartridge tailored for kick and close-miked bass amps (Shure PG52 spec sheet), with a published curve that humps at 60-100 Hz, dips broadly through 200-800 Hz, rises at 4-5 kHz and rolls off hard above 10 kHz. TalkBass rates it well above its price on bass cabinets specifically, because that curve is the right shape for one. Three gate calls here: B1 flat and the HPF at 100 because the 60-100 Hz hump is baked and the DI owns that lane anyway; B2 at 160 rather than the usual 300-400 because 200-800 Hz is a baked dip; B4 at 4500 trims the documented presence rise.

The HPF at 100 Hz is not a housekeeping filter — it is the lane boundary, drawn at the exact frequency the research named. Everything below belongs to the DI, everything above belongs to this mic, and the phase problem never arises. The 4.5 kHz trim takes off string and fret noise that reads as clank outdoors, and the 160 Hz cut is the KB's prescribed box position, sitting deliberately outside the capsule's own dip.

Ch 13 | Guitar · Sennheiser e609 Silver

`HPF 90 | LPF 12000 || B4 -4@4 kHz Q2 BELL   B3 -3@2.8 kHz Q2 BELL   B2 -7@300 Hz Q2 BELL   B1 flat`

40 Hz-15 kHz supercardioid with a presence peak at about 4 kHz and a broader midrange peak across 3-6 kHz (Sennheiser e609 Silver product specification / homestudiobasics). The KB independently flags it as edgy at 2.5-4 kHz on bright amps, and the neo-soul mixing guidance independently says to cut guitar at 2.5-3 kHz to stop it overlapping the vocal. Three sources, one move. Both B4 and B3 sit inside baked peaks, so both are trims.

There is one guitar in this band and it carries all the harmony, so it gets shaped rather than gutted: the vocal-clearing cut at 2.8 kHz is held to -3 even though the genre source and the outdoor rule would both support more. The 300 Hz cut, on the other hand, takes the full outdoor -7 without apology — a wooden cab in an open plaza is mechanical mud, which is precisely what the FSQ depth rule is for.

Ch 17 | Keys 1 · Whirlwind IMP (passive DI)

`HPF 45 | LPF 15000 || B4 -3@8 kHz Q2 BELL   B3 -4@2 kHz Q2 BELL   B2 -5@300 Hz Q1.8 BELL   B1 flat`

No capsule — a passive DI is neutral by definition, so every move here is a separation decision. The KB has no row for a keyboard or line-level board at all, the same gap logged on the 7/27 and 8/1 builds, so this unit is verdicted THIN and the values stay conservative. The fact doing the work comes from the genre research: neo-soul staging guidance is to reduce 8-10 kHz by 3-5 dB to fight stage reverb and to treat 500 Hz proximity warmth as a feature to protect, not mud to remove.

The 300 Hz cut runs -5, not the -8 the outdoor rule would give, and that is a deliberate trade with the reasoning on paper: the FSQ deep-cut rule exists to fight mud, and outdoors that mud is mechanical — box, cab, stage bleed — because an open plaza has no room buildup of its own. A Rhodes' low-mid warmth is the instrument, not mud. The 8 kHz trim is the genre source's own instruction and the humidity call reinforces it, so it survives twice over. 2 kHz comes off to keep the electric piano's bark out of the vocal.

Ch 18 | Keys 2 · Whirlwind IMP (passive DI)

`HPF 45 | LPF 15000 || B4 -3@8 kHz Q2 BELL   B3 -4@2 kHz Q2 BELL   B2 -5@300 Hz Q1.8 BELL   B1 flat`

Same passive DI, same THIN verdict, same genre-sourced reasoning as CH 17. The curves match on purpose: different EQ on the two halves of a stereo keyboard rig pulls the image off centre.

Identical to CH 17 by design. If it turns out these are two separate keyboards rather than one stereo rig, this is the channel to re-voice — move its 2 kHz cut to 1.2 kHz and leave CH 17 alone, and the two boards will separate without either losing its warmth.

HORNS

Ch 19 | Sax · Audio-Technica PRO 35

`HPF 120 | LPF 14000 || B4 -3@6 kHz Q2 BELL   B3 -5@2.5 kHz Q2 BELL   B2 -6@400 Hz Q2 BELL   B1 flat`

50 Hz-15 kHz with a built-in low-frequency roll-off at 80 Hz at 18 dB per octave, 145 dB max SPL and what Audio-Technica themselves call a subtly rounded treble (PRO35 product specification / Thomann). The KB adds the warning that matters on a close bell clip: plasticky upper-mid. The desk HPF sits at 120, above the capsule's own 80 Hz corner, so the two filters aren't duplicating each other — the desk one is there for key noise and stage rumble, not to re-cut what the mic already cut.

Alto sax fundamentals live between 150 and 700 Hz with presence at 2-4 kHz, so the -5 at 2500 is the primary move: it takes the honk out without touching the horn's body. In this band the sax sits inside the arrangement rather than over it, so nothing is added on top — and between the capsule's rounded treble and 87-94% humidity, an air lift would be the wrong call twice over. The 400 Hz cut at -6 handles the proximity box a bell clip generates at close range. COMP: Mustard Purple (optical), 3:1, 10 ms attack, 250 ms release, 3 dB of gain reduction — documented, not patched.

VOCALS

Ch 33 | Mic. Carr · Shure Beta 58A (Wireless 1)

`HPF 90 | LPF 15000 || B4 -3@10 kHz Q2 BELL   B3 -2@4 kHz Q3 BELL   B2 -5@700 Hz Q2 BELL   B1 flat`

50 Hz-16 kHz, attenuated below 500 Hz to counter proximity, with TWO presence peaks at 4 kHz and 10 kHz (Shure Beta 58A specification sheet; Sound On Sound Beta Series review) — reviewers regularly describe those peaks as grating. Supercardioid, so keep wedges 30-60 degrees off axis. Both B4 and B3 land on the measured peaks and both are cuts; there are no boosts on any vocal channel in this show. The template's baseline HPF of 184 Hz is overridden to 90 — 184 sits above the entire lower octave of any male voice, and E2 is 82 Hz.

Carr is the lead, the namesake and the reason the show is called The Shades, so this fader is built to stay the most intelligible voice on stage: the 4 kHz peak is only trimmed -2, on a tight Q3, where both harmony faders get -4 across a wider Q. Separation comes from the low-mid lane instead — his cut sits at 700 Hz while the harmonies are cut at 1500 and 1200, so when all three sing together no two cuts stack. That 700 Hz cut is held to -5 rather than the FSQ -8: this is a warm soul voice and outdoors there is no room buildup adding to his chest register. Writing B4 removes the template's -18 at 5 kHz Q20 feedback notch — ring this fader out at soundcheck. COMP: Mustard Purple (optical), 3:1, 10 ms attack, 200 ms release, 3-4 dB of gain reduction.

Ch 34 | Vox 2 · Shure Beta 58A (Wireless 2)

`HPF 110 | LPF 14000 || B4 -4@10 kHz Q2 BELL   B3 -4@4 kHz Q2 BELL   B2 -5@1.5 kHz Q2 BELL   B1 flat`

Same capsule and same two measured presence peaks at 4 kHz and 10 kHz as CH 33 (Shure spec sheet / SOS), trimmed harder here so this voice sits behind the lead rather than beside it. HPF at 110 rather than 90: 110 is the safe overlap of the male 60-100 Hz and female 80-120 Hz live-HPF ranges, which makes it correct whichever way this voice falls — the honest answer to a voice type nobody could confirm.

The band's own copy credits them with smooth harmonies, so these two faders have to blend with each other and stay out of Carr's way. Both peak trims run -4 to place this voice behind the lead, and the -5 at 1500 Hz is this fader's private nasal lane — nobody else is cut there. Held at -5 rather than -8 for the same reason as the lead: a harmony voice outdoors has no room buildup behind it.

Ch 35 | Vox 3 · Shure Beta 58A (Wireless 3)

`HPF 110 | LPF 14000 || B4 -4@10 kHz Q2 BELL   B3 -4@4 kHz Q2 BELL   B2 -5@1.2 kHz Q2 BELL   B1 flat`

Identical capsule treatment to CH 34 — the same two measured peaks trimmed -4 each, the same male-and-female-safe HPF at 110.

The third voice, and the only thing separating it from CH 34 is the one band that needs to be different: its nasal cut sits at 1200 Hz where CH 34's sits at 1500. That is deliberate and it is the whole point of slotting three vocals — with cuts-only vocals, separation has to come from WHERE each voice is cut, and three identical curves under a note saying 'blend the harmonies' would be a failed build.
